

eqannotate

Declarative equation annotations with automatic layout

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Abstract

`eqannotate` is a LaTeX package for labelled display mathematics. Mark a term, declare its meaning, and let the package arrange the annotation. This manual documents the stable v0.1.1 user interface.

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1 Introduction

EqAnnotate is for *display* equations. Its normal workflow is: mark a term, declare a label, then compile to convergence. It is automatic first, with a manual fallback for exceptional formulas. No shell escape, Python, network access, or external service is needed.

2 Installation

Place `eqannotate.sty` beside the main document and write:

```
\usepackage{eqannotate}
```

For a user TeX tree, install it below `tex/latex/eqannotate/`. Requirements are LaTeX2e, `amsmath`, `xcolor`, TikZ with `tikzmark`, `arrows.meta`, and `calc`, plus `expl3/xparse`. The tested engines are pdfLaTeX, LuaLaTeX, and XeLaTeX.

3 Quick start

$$p(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right)$$

Mark identifiers are local to one display. The declarations may follow the formula in any order.

4 Core API

4.1 Marking terms

```
\eqmark[<color>]{<id>}{<math>}
```

The optional color is used by the colorful theme. IDs are semantic names local to one wrapper. Duplicate marks keep the first target and issue a warning.

4.2 Automatic labels

```
\eqannotate[prefer=auto|above|below]{<id>}{<label>}
```

The optional `prefer` value is a soft side preference, not a coordinate. A label with no matching mark is skipped with a warning.

5 Display environments

Four wrappers are available:

annotatedequation A single display.

annotatedalign A multi-row aligned display.

annotatedgather A centered multi-row display.

annotatedmultline A long display split across rows.

They are unnumbered by default. Adding `[numbered]` gives the complete wrapper one outer equation number; ordinary `\label` and `\ref` work as usual.

$$\begin{array}{l}
 \text{Initial residual} \\
 | \\
 r_0 = x_0 - y, \\
 r_{k+1} = r_k - \alpha_k A p_k, \\
 x_{k+1} = x_k + \alpha_k p_k \\
 \text{State update} \quad \text{Residual correction}
 \end{array} \tag{1}$$

Equation 1 is one numbered aligned block.

$$\begin{array}{l}
 \text{Energy objective} \\
 | \\
 E(x) = D(x) + R(x), \\
 \nabla E(x) = \nabla D(x) + \nabla R(x) \\
 \text{Objective gradient}
 \end{array}$$

$$\begin{array}{l}
 \text{Reconstruction term} \quad \text{Adversarial term} \\
 | \quad | \\
 \mathcal{L}(\theta) = \lambda_1 \mathcal{L}_{\text{rec}} + \lambda_2 \mathcal{L}_{\text{adv}} \\
 \quad \quad \quad + \lambda_3 \mathcal{L}_{\text{cyc}} + \lambda_4 \mathcal{R}(\theta) \\
 \quad \quad \quad \quad \quad \quad \text{Prior regularizer}
 \end{array}$$

The multiline wrapper retains multiline-style first/last-row alignment. It also supports the native commands `\shoveleft` and `\shoveright`.

6 Automatic layout

Automatic placement chooses sides, wraps long labels, performs bounded horizontal de-overlap, allocates lanes, routes connectors, observes the active local `\linewidth`, and reserves vertical space. These are layout strategies, not an unconditional promise that every imaginable document layout can be solved without intervention.

$$\mathcal{J} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{reg}} + \gamma \mathcal{L}_{\text{aux}} + \rho \mathcal{R}(\theta)$$

Data fidelity Auxiliary objective
Regularization Prior term

7 Themes and callout styles

```
\eqannotatecolortheme{colorful} % or mono
\eqannotatecalloutstyle{leader} % or arrow
```

In `colorful`, marked terms have pastel backgrounds and related labels and connectors share their colors. In `mono`, term backgrounds are removed and callouts are black. A `leader` is a plain connector; an `arrow` points toward the target.

8 Manual fallback

```
\eqannotatemanual[<label TikZ options>][<to-path options>]
  {<id>}{<label>}
```

Use manual placement only when a formula needs exact control:

$$y = f_{\theta}(x) + \lambda g(x)$$

Base model Manually positioned correction

Manual labels retain the theme, callout style, masking, and vertical reservation, but bypass automatic collision avoidance, column clamping, lane assignment, and crossing optimization for that label. A manual declaration wins over an automatic declaration for the same ID and emits a warning.

9 Numbering and references

Each numbered wrapper has one outer number. Native per-row numbering for aligned/gathered wrappers and arbitrary custom `\tag` support are outside v0.1. Place `\label` inside a numbered wrapper and reference it normally.

10 Compilation and convergence

Use a normal workflow such as:

```
latexmk -pdf main.tex
```

When compiling by hand, rerun while the log reports

```
Rerun LaTeX for optimized annotation spacing.
```

The package uses remembered positions and the `.aux` file, so a fresh document commonly needs multiple passes.

11 Diagnostics

Diagnostics cover a missing mark, duplicate mark, duplicate annotation, automatic/manual duplication, and commands used outside an EqAnnotate display. The last case is an explicit package error; the other authoring mistakes have readable warning behavior.

12 Known limitations and v0.1 boundaries

EqAnnotate does not annotate inline mathematics. v0.1 has no per-row numbering for multi-line wrappers, arbitrary `\tag`, or `\intertext`/`\shortintertext` inside its wrappers. On a non-white page, set:

```
\renewcommand\eqannotatebackgroundcolor{<color>}
```

Automatic page-background detection is not provided.

13 Complete examples

The distribution includes the following runnable public-interface examples:

- `examples/basic.tex`
- `examples/style-gallery.tex`
- `examples/amsmath-gallery.tex`
- `examples/manual-gallery.tex`

They illustrate use of the public interface, not implementation knowledge.

14 License and project links

EqAnnotate is available under the MIT License. The [GitHub repository](#) hosts source and releases; report problems through the [issue tracker](#).